METRIC

DOD-W-63546A
W/AMENDMENT 1
12 April 2005

MILITARY SPECIFICATION

WASHER, KEY, METRIC
GENERAL SPECIFICATION FOR

Inactive for new design after 1 October 1999

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers flat long, flat long and short, and external types of washer, key, metric.

2. APPLICABLE DOCUMENTS

Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-495 - Finish, Chemical Black, For Copper Alloys

(Copies of specifications, standards, handbooks, drawing, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

(See supplement 1 for list of associated specifications.)

Comments, suggestions, or questions on this document should be addressed to: Defense Supply Center Philadelphia, ATTN: DSCP-ITAA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or emailed to dscpg&ispeccomments@dla.mil. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at http://assist.daps.dla.mil/

AMSC N/A FSC 5310

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

American Society For Testing And Materials Standard (ASTM)

ASTM A109	- Standard Specification for Steel, Strip, Carbon (0.25 Maximum Percent), Cold-Rolled
ASTM A153	- Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A342	- Standard Test Methods for Permeability of Feebly Magnetic Materials
ASTM B36	- Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B633	- Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM E1444	- Standard Practice for Magnetic Particle Examination

(Application for copies should be addressed to ASTM, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

American Society For Quality Control:

ASQ Z1.4 -Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society For Quality Control, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

Society of Automotive Engineer (SAE):

AMS-STD-66	-Steel: Chemical Composition and Hardenability
AMS-QQ-P-416	-Plating, Cadmium (Electrodeposited)

(Application for copies should be addressed to the SAE, International, 400 Commonwealth Drive, Warrendale. PA 15096-0001.)

Aerospace Industries Association (AIA)

NASM 1312 - Fastener Test Methods

(Application for copies of NASC publications should be addressed to the Aerospace Industries Association, 1250 Eye Street NW, Washington DC 20005.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. Recycled and reclaimed materials (see 3.1) shall be used to the maximum extent practicable.

3.2.1 Brass. Brass washers shall conform to the requirements of ASTM B36 copper alloy UNS C26800, temper half-hard.

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3.2.2 Steel. Carbon steel washers shall be of composition 1008 to 1018, cold rolled, no. 4 temper, no. 3 finish in accordance with ASTM A109.

3.3 Hardness.

3.3.1 Carbon steel washers. Carbon steel washers shall have a maximum hardness of 65 HRB. Washers having a thickness less than .75 mm shall have a maximum hardness of 60 HR30T.

3.3.2 Brass washers. Brass washers shall have a maximum of 70 HRB. Washers having a thickness less than .75 mm shall have a maximum hardness of 65 HR30T.

3.4 Finish. The protective finish of the washers when specified in the applicable specification sheet shall be as follows (see 4.3.3).

3.4.1 Black chemical. Brass washers shall be black oxide coated in accordance with MIL-F-495.

3.4.2 Cadmium plate. Carbon steel washers shall be cadmium plated in accordance with SAE-AMS-QQ-P-416, type II, class 2. The coating thickness shall be 7.6 µm (micrometers).

3.4.3 Zinc coating. Carbon steel washers shall be zinc coated in accordance with ASTM B633, Fe/Zn 5, type II.

3.4.4 Galvanized zinc. (hot-dipped). Carbon steel washers shall be hot-dipped galvanized in accordance with ASTM A153.

3.5 Dimensions. Dimensions and tolerances shall be in accordance with the applicable specification sheet and as specified herein.

3.5.1 Parallelism. Washer faces shall be parallel within:

25 µm for thickness less than .50 mm
50 µm for thickness .50 mm to 1.00 mm
100 µm for thickness 1.00 mm to 2.50 mm
150 µm for thickness 2.50 mm to 4.00 mm

3.5.1 Flatness. Washer faces shall be flat within:

175 µm for hole sizes thru 13 mm
250 µm for hole sizes 14 mm thru 34 mm
380 µm for hole sizes over 34 mm

3.5.3 Concentricity. The outside diameter (OD) of the washer shall be concentric with the inside diameter (ID) within the following limits as specified in Table I.

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TABLE I. Concentricity.

Outside diameter (OD) mm		Concentricity µm
Over	To	
1	3	125
3	6	150
6	10	180
10	18	215
18	30	260
30	50	310
50	80	370
80	120	435

3.6 Cracks. Washers shall be free from cracks, flaws, and pits in any location. A crack is a clean crystalline fracture passing through or across the grain boundaries without inclusion of foreign elements.

3.7 Workmanship. Washers shall be free from surface contamination tool marks and other imperfections which may adversely affect useability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall be as specified in table II.

TABLE II. Quality conformance inspections.

Inspections	Requirement Paragraph	Test method Paragraph
Dimensions	3.5	4.3.1
Magnetic particle (cracks)	3.6	4.3.2
Protective finish	3.4	4.3.3
Hardness	3.3.1, 3.3.2	4.3.4

4.2.1 Inspection lot. An inspection lot shall consist of completed washers covered by a single specification sheet, produced under essentially the same conditions and offered for inspection at one time.

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4.2.2 Rejected lot. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lot shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.2.3 Sampling for visual and dimensional examination. Sampling shall be in accordance with inspection level S-3 of ASQ Z1.4 with an acceptable quality level (AQL) of 2.5 percent defective for major defects. Minor defects shall be in accordance with inspection level S-3 of ASQ Z1.4 with AQL of 4.0 percent defective.

4.2.4 Sampling for finish test. Sampling for protective finish shall be in accordance with the applicable specifications referenced in 3.4.1 thru 3.4.4.

4.2.5 Sampling for other tests. Sampling for magnetic particle (cracks), and hardness inspection shall be in accordance with inspection level S-1 of ASQ Z1.4 with an AQL of 1.5 percent defective.

4.3 Methods of Inspection.

4.3.1 Visual and dimensional examination. Samples taken as specified in 4.2.3 shall be thoroughly examined to determine conformance with this specification and applicable specification sheet. Examination shall be conduced in accordance with table III.

TABLE III. Classification of defects.

Category	Defect	Inspection Method
Critical	None defined	
Major		
101	Inside diameter, not as specified (see 3.5)	SIE 1/
102	Outside diameter, not as specified (see 3.5)	SIE
103	Thickness, not as specified (see 3.5)	SIE
104	Parallelism, not as specified (see 3.5.1)	SIE
105	Flatness, not as specified (see 3.5.2)	SIE
106	Evidence of cracks, flaws or pits (see 3.6)	Visual
Minor		
201	Protective finish, missing or incomplete (see 3.4)	Visual
202	Other dimensions, not as specified (see 3.5)	SIE
203	Workmanship (see 3.7)	Visual

1/ Standard inspection equipment.

4.3.2 Magnetic particle (cracks) inspection. When specified (see 6.2), samples taken as specified in 4.2.5 shall be subjected to magnetic particle inspection in accordance with ASTM E1444. There shall be no evidence of cracks or pits as specified in 3.6.

4.3.3 Finish test. Carbon steel washers shall conform to the finish test of SAE-AMS-QQ-P-416, ASTM B633 or ASTM A153 as specified in 3.4. Brass washers shall conform to the finish test of MIL-F-495.

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4.3.4 Hardness test. Washers shall be tested for surface hardness in accordance with NASM1312, test method 6.

4.3.5 Material inspection. Material inspection shall consist of certification supported verifying data that the materials used in fabricating washers are in accordance with the applicable paragraphs listed in 3.2.1, 3.2.2, 3.3.1 and 3.3.2.

4.3.6 Deleted.

5. PACKAGING

5.1 Packaging. For Acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contracting the responsible packaging activity.

6. NOTES

6.1 Intended use. This specification covers various types of washers, key, flat, for general purpose, aircraft and general metric bolt and nut application. Their purpose is to provide an increase bearing surface and to prevent galling of the base material as well as safetying metric fasteners.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and applicable specification sheet
b. Applicable specification sheet part number
c. Magnetic particle (see 4.3.2)

6.3 Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army – AR
Navy – SH
Air Force – 99

Review activities:
Army – AT, AV, MI, CR4
Air Force – 71

Preparing activity:
DLA-IS
(Project 5310-2719)

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